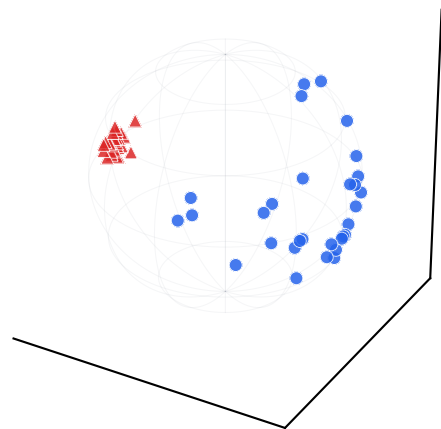
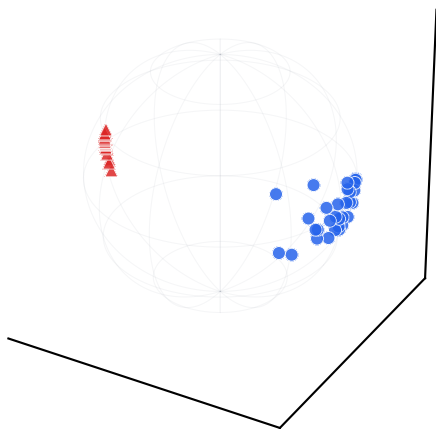


Unit Normalized Direction Vectors on S² (known (H) vs unknown (TOFU))
 $\mathbf{d}_b = \text{mean}_{\text{batch}}(\mathbf{h}_{\text{reg}} - \mathbf{h}_{\text{anon}}) / \|\cdot\|$ projected via PCA(3D) → L2-normalize.
 HP clusters at a pole (coherent); TOFU scatters uniformly (fragmented).

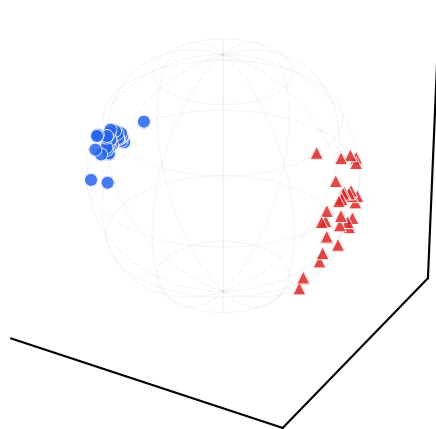
Qwen-7B
(layer 0)



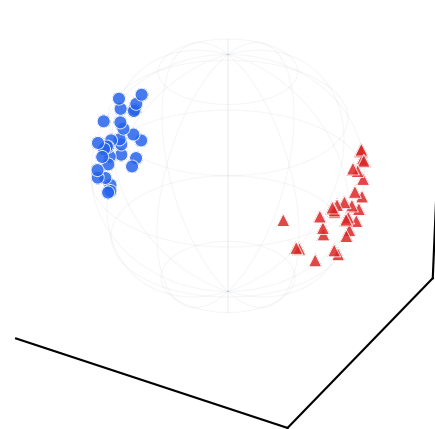
Qwen-14B
(layer 17)



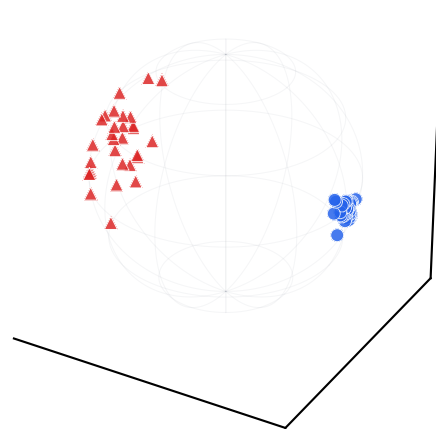
Qwen2.5-7B
(layer 27)



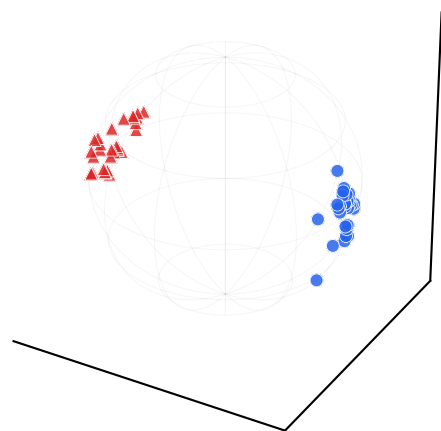
Qwen2.5-14B
(layer 47)



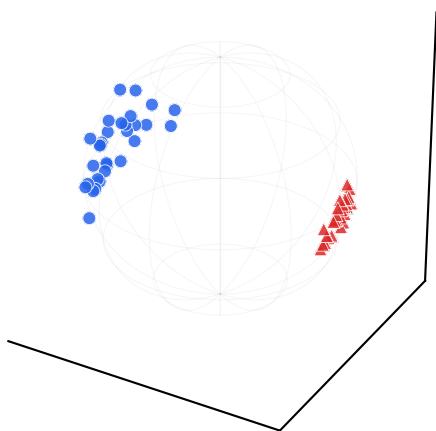
Gemma-2B
(layer 18)



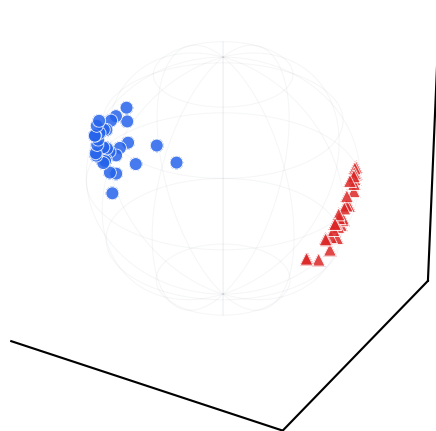
Gemma-7B
(layer 26)



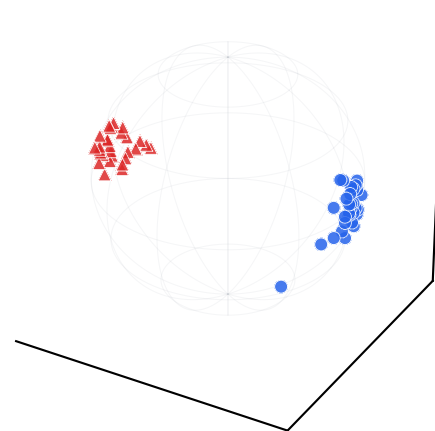
Llama-2-7B
(layer 31)



Llama-2-13B
(layer 28)



Llama-3-8B
(layer 25)



Llama-3.1-8B
(layer 25)

